

Practice Problems Sheet VII

(Inclusion-Exclusion)

- Q1** Find the number of positive integers not exceeding 100 that are odd or square or an integer.
- Q2** Find the number of solutions of the equation $x_1 + x_2 + x_3 + x_4 = 17$, where x_i 's are non-negative integers, such that $x_1 \leq 3$, $x_2 \leq 4$, $x_3 \leq 5$, and $x_4 \leq 8$.
- Q3** Find the number of primes less than 200 using the principle of inclusion-exclusion.
- Q4** An integer is called *squarefree* if it is not divisible by the square of a positive integer > 1 . Find the number of *squarefree* positive integers less than 100.
- Q5** How many integers in interval $(1, 10000)$ are a second or higher power of some integer?
- Q6** In how many ways can digits $0, 1, 2, \dots, 8, 9$ be arranged so that no even digit is in its original place?
- Q7** Let D_n be the number of derangements of n objects.
- (a) Using the result $D_n = (n-1)(D_{n-1} + D_{n-2})$, for $n \geq 2$, show that $D_n = nD_{n-1} + (-1)^n$, for $n \geq 1$.
 - (b) Using part (a), derive the explicit formula for D_n .
- Q8** Show that if n is a positive integer, then

$$n! = \sum_{i=0}^n \binom{n}{i} D_i$$

End